Unit: undefined
Topic: Algebra 1: Polynomials

Name: ____________
Date: ________

Level 1: Basic

Q1. What is the degree of the polynomial $3x^2 + 2x - 4$?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

Q2. What is the coefficient of x in the polynomial $2x^2 + 5x - 1$?
(A) 1
(B) 2
(C) 3
(D) 5
(E) 6

Q3. Simplify the expression $2x + 3x$ to get?
(A) $5x$
(B) $6x$
(C) $4x$
(D) x
(E) $3x$

Q4. What is the constant term in the polynomial $x^2 + 2x + 1$?
(A) 0
(B) 1
(C) 2
(D) 3
(E) 4

Q5. What is the degree of the polynomial $x^3 - 2x^2 + x - 1$?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

Q6. Identify the polynomial $x^2 + 2x + 1$ as?
(A) Monomial
(B) Binomial
(C) Trinomial
(D) Quadrinomial
(E) None

Q7. What is the leading coefficient of the polynomial $3x^2 - 2x - 4$?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

Q8. Combine like terms: $2x^2 + 3x + x^2 + 2x$?
(A) $3x^2 + 5x$
(B) $4x^2 + x$
(C) $x^2 + 2x$
(D) $3x^2 + x$
(E) $x^2 + 5x$

Open Ended Questions (FRQ)

Q9. Explain the concept of the degree of a polynomial and provide an example with the polynomial $x^2 + 2x - 3$.

Q10. Draw a diagram that represents the polynomial x^2 and explain its characteristics.

Q11. Explain the concept of combining like terms in a polynomial and provide an example with the expression $2x + 3x - x$.

Q12. Describe the relationship between the coefficients and the terms of a polynomial, and provide an example with the polynomial $2x^2 + 3x - 4$.

Chef's Tips

- For Level 1: Basic, focus on identifying and understanding basic concepts of polynomials such as degree, coefficients, and combining like terms.
- Use visual aids and diagrams to help students understand complex concepts.
- Encourage students to provide explanations and examples to demonstrate their understanding of polynomial concepts.